(a) Experience guides when Astra reviews

Conceptual task illustration

Task history Target

Task instruction
Pick up the highlighted block and place it in the target area.

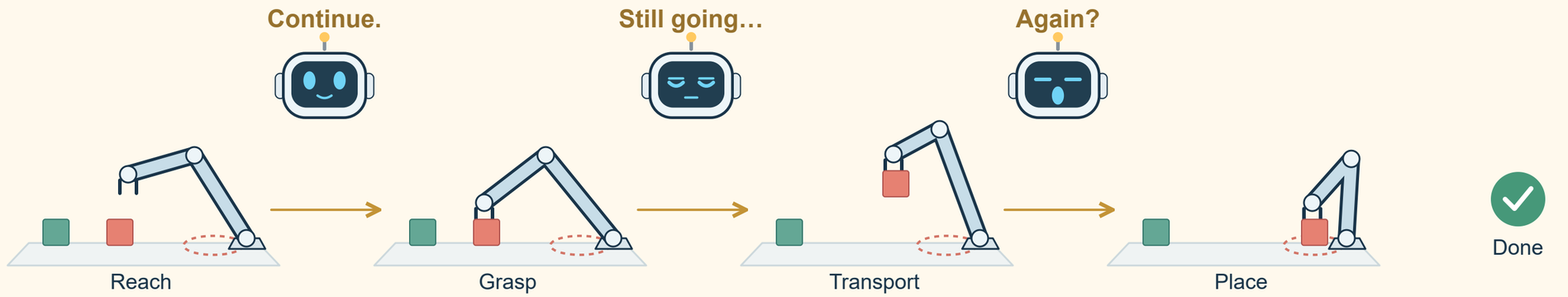
Astra

Grounded subgoal
Place the red block in the target area.

Per-chunk review

Same subgoal.
Another check.
Another “continue”.

Repeated Astra inference



Our harness

Experience
Routine transport
→ longer budget
→ selected monitor
Read · reflect · revise

VLA weights fixed

Budget: up to 100 control steps
Monitor: gripper change

Illustrative

Astra

Zzz
On standby

Monitor stays active
Event → early review

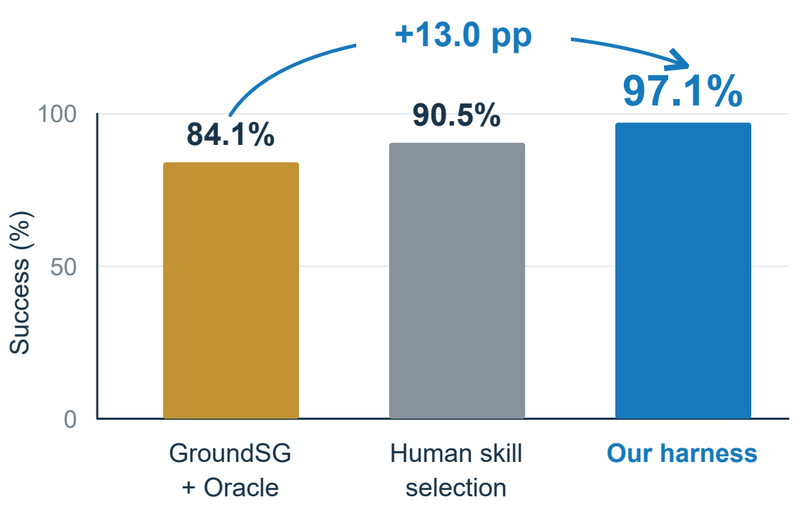
Review
Correct if needed

Budget reached

VLA action chunks · fresh observations throughout

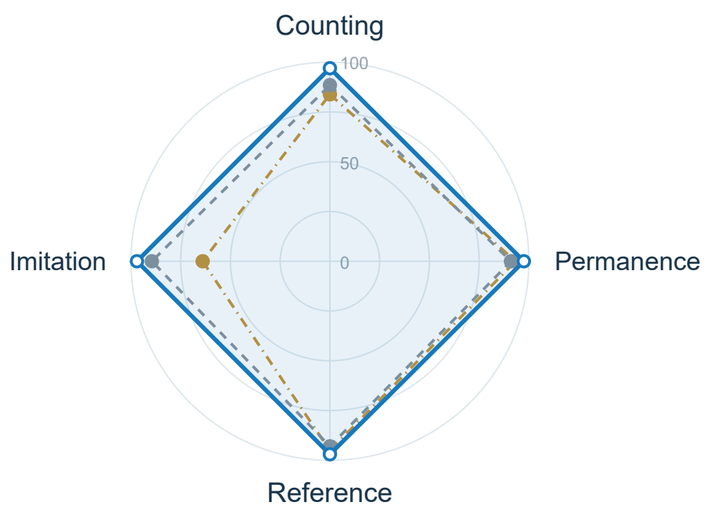
Outcome & reflection → revise for later episodes

(b) RoboMME Performance



Ours: 16 tasks · 800 episodes

(c) Four memory suites



≥97% in every suite

(d) Less repeated review

54.9%

of control steps without additional **Astra review**



Across 800 episodes

GroundSG + Oracle · Human skill selection · Our harness

Published reference scores; broader comparisons in the main results.